

Characterization of nLGAD sensors from FBK using TCT techniques

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Abstract—nLGADs were proposed as detectors for detecting low-penetrating particles, such as soft X-rays, ultraviolet light, or low-energy charged particles. The development of this type of device can be a key factor for developing next generation silicon detectors introduced into future experiments like the FCC. However, the best design of these types of devices is still under research. In this work multiple detectors from different wafers are studied. A deep study of these types of detectors is shown in order to give an explanation on the induced signals from different scenarios, to figure out the variations on the gain reduction mechanism presented in nLGADs and how the composition of the n -type gain layer can affect this phenomenon.

Index Terms—nLGAD, TPA-TCT, SPA-TCT, Gain

I. INTRODUCTION

Low Gain Avalanche Detectors (LGADs) were introduced as an improvement over the conventional PiN (p-intrinsic-n) diodes for high-energy physics experiments, where electronics processing is essential. Both PiNs and LGADs suffer from significant signal degradation after being irradiated [1]. However, LGADs introduce a gain mechanism which enhance the signal amplitude throughout controlled avalanche electrons multiplication. This gain mechanism is crucial for maintaining the integrity of the signal up to significant radiation levels, but also for achieving good time resolution.

To extend the capabilities of this technology, n -type LGADs (referred to as nLGADs from this point on) have been developed. Proposed by the universities and Instituto Nazionale di Fisica Nucleare (INFN) sections of Trento, Torino, Perugia, and the Fondazione Bruno Kessler (FBK), nLGADs are primarily investigated for two main purposes: testing their performance for low-penetrating particles and studying the donor removal mechanism in n -type gain layers.

Regarding to their performance, conventional LGADs exhibit poor detection efficiency for low-penetrating particles in silicon, such as soft X-rays photons or low-energy charged ions [2-5]. This is primarily because standard LGADs feature a relatively thick top n^+ layer, resulting in a larger undepleted volume where shallow particles are absorbed without generating a measurable signal. The p-in-n design of nLGADs overcomes this limitation. In these devices, the top p^+ layer is almost completely depleted, drastically reducing the dead material and minimizing the loss of absorbed photons. Furthermore, nLGADs include an n -type gain layer which enables the

multiplication mechanism as in standard LGADs. In this architecture, the electron drive the avalanche that makes the multiplication process easier to control [6, 7].

Beyond the detection of low-penetrating particles, nLGADs are highly interesting for studying the donor removal mechanism in n -type gain implants. Understanding this phenomenon is essential for the future production of compensated LGADs, which are currently being developed within the framework of the EXFLU and COMPLEX projects [8, 9]. A compensated LGAD implements both acceptor and donor gain implants [10], so that the effective doping concentration is less affected by radiation and results in more radiation resistant detectors useful for future experiments, such as FCC-hh, where radiation damage and the timing resolutions and signal strength are crucial [11]. Developing nLGADs will provide very useful information to optimize this design, resulting in highly radiation-resistant detectors necessary for future colliders.

II. DEVICES UNDER TEST

The studied detectors were manufactured at Fondazione Bruno Kessler (FBK) on a 55 μm thick high resistivity ($> 10 \text{ k}\Omega \cdot \text{cm}$) active layer on top of a phosphorus-doped substrate [10]. Below the active region, a support wafer of $\sim 520 \mu\text{m}$ thickness, made of low resistivity material ($\mathcal{O}(10 \Omega \cdot \text{cm})$) is used to provide mechanical support to the thin active layer. During this fabrication three different p^{++} ohmic contact designs were produced, having a shallow, deep and shallow+deep p^{++} profile for the different wafers. At the same time, another three different designs were produced for the n^+ gain implants in which two phosphorus (shallow and deep) and one arsenic shallow implant are present. For a complete description of the production and its split table, refer to [10]. In this study, we are concentrating on the *phosphorous-deep* and *arsenic-shallow* contact types. Tab I summarizes the characteristics of the Devices Under Test (DUTs) employed in this work: the type of p^{++} ohmic contact, the shape of the gain layer and its dose.

On the fabricated wafers, nLGADs and p -in- n diodes (referred to PiNs from this point on) are present. The studied detectors have a square entrance active area of $1.205 \times 1.205 \text{ mm}^2$, which allow to use the Transient Current Technique (TCT) measurement. For the nLGADs, the gain layer is present, while for the PiNs, the gain layer is not.

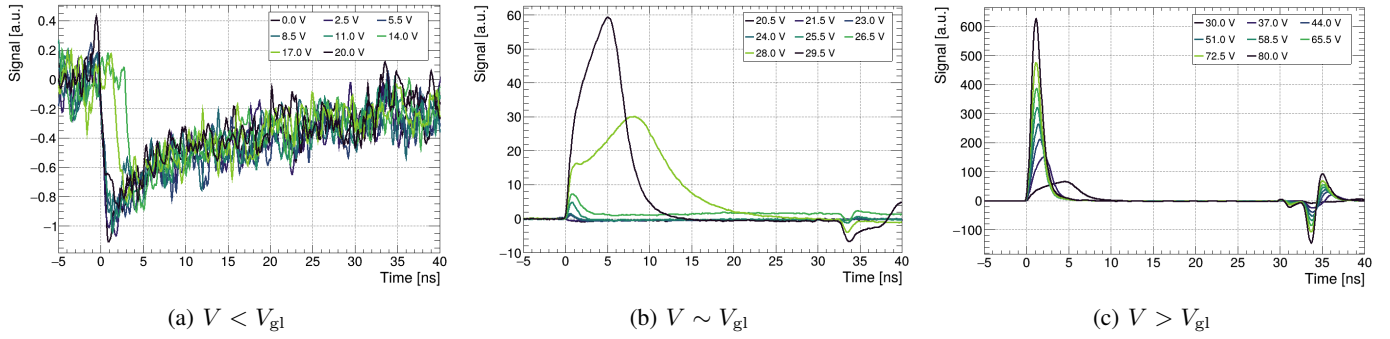


Figure 1: TCT waveforms acquired at various bias voltages, illustrating the three considered scenarios of the nLGAD W6 sensor: (a) Below the gain layer depletion voltage ($V < V_{gl}$), showing pure drift without multiplication; (b) Around V_{gl} ($V \sim V_{gl}$), showing the onset of the avalanche process; and (c) Above V_{gl} ($V > V_{gl}$), where significant charge multiplication occurs.

Table I: Studied wafer characteristics from FBK. For each type of wafer, the used contact, gain type and the amount of gain dose used for the gain layer is shown.

Wafer	Contact	Gain	Gain Dose
6	Shallow + Deep	Phosphorous Deep	1.30
11	Shallow	Phosphorous Deep	0.96
12	Shallow	Phosphorous Deep	1.00
13	Shallow	Arsenic Shallow	1.30

III. EXPERIMENTAL SETUP AT CERN

To carry out the characterization of these detectors, the Transient Current Technique (TCT) setup of the Solid State Detectors group laboratory at the EP-DT department from CERN was used. This type of technique is based on the analysis of transient current pulses, originating from the drift of charge carriers in the electric field inside a biased sensor [12–14]. Out of the TCT available techniques, in this study Single Photon Absorption (SPA-TCT) and Two Photon Absorption (TPA-TCT) has been exploited. All studied devices were mounted on a passive PCB. The bias was always applied from the p^{++} side of the detector and the signal readout is achieved thanks to the internal bias tee contained in the employed Cividec 2 GHz 20 dB amplifier.

A. Single Photon Absorption (SPA)

This technique uses photons of the energy of silicon bandgap to excite electrons from the valence band to the conduction band, creating an electron-hole pair. Due to the external applied bias voltage, this pair will drift towards the anode and the cathode of the electric circuit, inducing a signal that can be analyzed. In the TCT setup for SPA, two different types of lasers were used: a red laser with $\lambda_{red} \approx 660$ nm and an infrared laser with $\lambda_{IR} \approx 1064$ nm. These two wavelengths provide different information about the induced signal, useful to understand both the charge carriers' drift inside the devices and the behavior of these detectors when using photons as minimum ionizing particles (MIPs) [13].

B. Two Photon Absorption (TPA)

In conventional SPA-TCT measurements, if the energy of a photon is less than the bandgap energy, no signal is produced. However, if a highly focused femtosecond laser provides a sufficiently high photon density, the simultaneous absorption of two sub-bandgap photons becomes highly probable via a virtual state [15, 16]. TPA-TCT allows the absorption of photons with energies smaller than the bandgap energy of silicon, which is possible for wavelengths between 1100 nm and 2300 nm [14]. During this work, a wavelength of $\lambda_{TPA} \approx 1550$ nm is used for analyzing the induced current signals [15]. In contrast with SPA, TPA enables to study devices with three dimensional spatial resolution.

IV. RESULTS

All the results presented in this report have been obtained using non irradiated detectors. This study provides an overview about how these types of devices can perform before they are damaged by radiation, giving an understanding about the different n -type gain layer designs for improving their design in future productions.

A. Internal electric field studies

Fig. 1 presents the different waveforms waveforms acquired at a reverse bias lower, equal and higher than the gain layer depletion voltage applied to the W6 nLGAD during a measurement in SPA using red light. For bias voltage above the gain layer depletion voltage V_{gl} , the waveforms presents a normal and expected behavior when shooting the laser on the top of the detector (Fig. 1c). However, if the applied bias voltage starts to decrease, smaller and larger tails start to appear (Fig. 1b). These long tails are present for voltages above the gain layer depletion voltage, which can be explained by the slow diffusion of the gain electrons from the gain layer until they are collected. Finally, when the external applied voltage becomes smaller than the V_{gl} , a negative signal starts to appear, even though a very small positive signal is present at the very beginning of the signal, indicating that carriers are being collected in the opposite direction to the external applied electric field. This type of signal is also presented in the rest

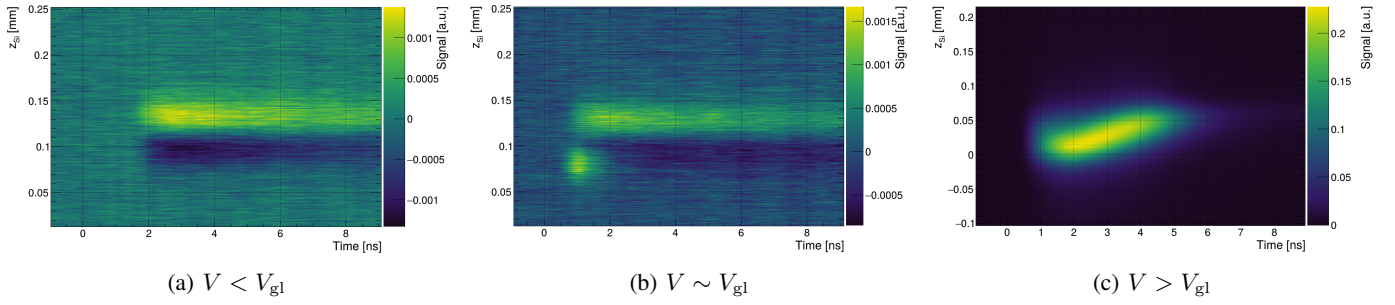


Figure 2: 2D plots showing the evolution of the signal waveform for each depth position (z_{Si}) inside the detector. Time is represented on the horizontal axis, the z -coordinate on the vertical axis, and the induced signal amplitude by the color scale, capturing the distinct charge collection dynamics for (a) $V < V_{gl}$, (b) $V \sim V_{gl}$, and (c) $V > V_{gl}$

of the nLGAD detectors. To avoid redundancy, the following plots will refer exclusively to the aforementioned W6 nLGAD, since identical behavior was observed across all other devices.

In order to understand better the structure and the physical properties for these specific scenarios, TPA measurements were performed by measuring the different induced signals for different positions of the focal coordinate (z). In the plots shown in Fig. 2 a two dimensional representation of the waveforms as a function of the z coordinate is shown, where the values near zero represent the beginning of the detector. Note that in these representations the z_{Si} is referred to the real position of the laser, which differs from the position of the detectors, due to the refraction of the laser in silicon, multiplying the z value from the stage by a constant factor (~ 4) [13]. For an applied voltage above V_{gl} (Fig. 2c), a signal coherent with the Shockley-Ramo theorem develops along the entire depleted volume of the device. Under these conditions, the sensor behaves as expected, making it possible to estimate the device's sensitive thickness. However, as the bias voltage decreases, some of the characteristics presented in the waveforms show their effect. For the $V \sim V_{gl}$ scenario, a small positive signal is induced in the inner part of the gain layer ($0.06\mu\text{m} < z < 0.09\mu\text{m}$), followed by a negative induced signal. Simultaneously, positive induced signals are collected in the back part of the detector ($z > 0.1\mu\text{m}$), associated to the support wafer (Fig. 2b). The presence of the small induced signal in this scenario can be associated to partial depletion of the n -type gain layer. Finally, for the $V < V_{gl}$ scenario, complete negative pulses are induced inside the active region of the detector, while positive pulses in the region of the support wafer. In contrast to the previous scenario, the depletion of the gain layer is not present, which results in no positive pulses.

To have a complete understanding of the physical behavior of the carriers within the detector, the collected charge and drift velocities of the carriers at different positions in z for a set of bias voltages were studied. The collected charge gives information about how charges move inside the detector, while an analysis of the drift velocity provides information about the internal electric field in the device at different points by following the Shockley-Ramo theorem [17]. This connection

is established by the fundamental relation:

$$v_{\text{drift}} = \mu(E)E,$$

where $\mu(E)$ is the field-dependent mobility of the charge carriers. Consequently, mapping the velocity of the charges allows for a direct evaluation of the electric field distribution inside the sensor. For this study, the collected charge was obtained by integrating the waveforms within 30 ns, because reflections of the signal inside cables appear at later times. For drift velocity estimation, the initial 0.6 ns were integrated, since for short windows $Q(\Delta t) \propto v_{\text{drift}}\Delta t$. The results were studied in the same three scenarios as in previous cases, which can be seen in Fig. 3 for the collected charge and in Fig. 4 for the drift velocities. Furthermore, as observed in the high-voltage regimes, the drift velocities approach a plateau. This behavior is expected, since charge carriers in silicon reach a saturation velocity when subjected to the strong electric fields present at higher biases.

Figs. 3 and 4 show an evolution of both collected charge and drift velocity through different z positions of the devices over a set of bias voltages. For the case below the V_{gl} , not only negative collected charges, but negative drift velocities within the active region of the detector are shown, indicating that the carriers drift in the opposite direction of the external electric field, while inside the support wafer, charged carriers go in the direction of the external field. At bias voltages close to the V_{gl} , the positive movement of the carriers persist in the support wafer. However, the increase is demonstrated in both the collected charge and the drift velocity, confirming the partial depletion of the gain layer. In this scenario we start to see positive signals as shown in Fig. 2c, meaning that the charge carriers drift in the same direction of the external field.

B. Thickness estimation

As bias voltages were increased beyond the full depletion voltage, the collected charge profiles exhibited a spatial broadening along the z -axis, as shown in Fig. 3c. The study of these charge-versus- z profiles made it possible to extrapolate the actual depleted thickness of the sensors, defining the effective region useful for particle detection. To extract this active thickness (Δz), the data were fitted with the convolution of a Lorentzian

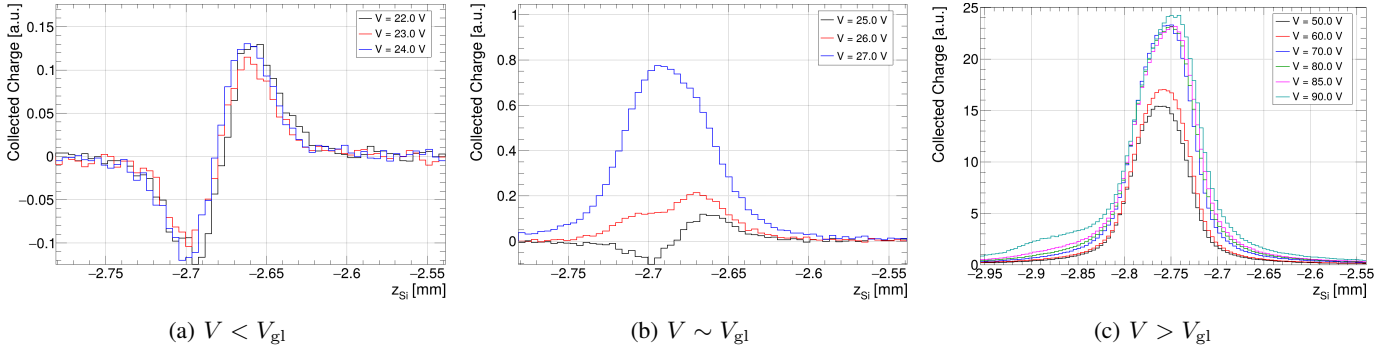


Figure 3: Total collected charge profiles as a function of the focal point depth (z_{Si}) using the TPA-TCT setup. The profiles show the spatial dependence of charge collection and the significant increase in signal amplitude at voltages well above V_{g1} .

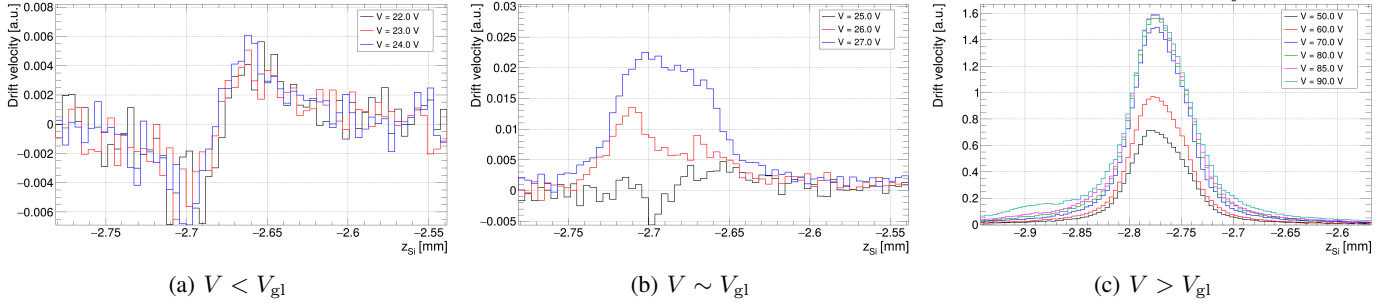


Figure 4: Spatial profiles of the estimated drift velocity along the sensor depth (z_{Si}), obtained by integrating the initial 0.6 ns of the TPA-TCT waveforms. The profiles show the electric field distribution (a) before the full depletion of the gain layer, (b) during the gain layer depletion and (c) for high bias voltages after the gain layer depletion.

V_{bias} [V]	50	60	70	80	90
Δz [μm]	61.38	62.26	63.41	65.56	66.77
$\delta\Delta z$ [μm]	0.31	0.38	0.45	0.55	0.95

Table II: Extrapolated active thickness (Δz) of the detector at different applied bias voltages. The errors ($\delta\Delta z$) represent the statistical uncertainties extracted from the fit.

and a box function [18] (see the Appendix). This specific model accounts for the Lorentzian spatial distribution of the TPA laser energy deposition and the intrinsic rectangular geometry of the detector's active volume. An example of this analysis is shown in Fig. 5, where the fit is applied to the collected charge profile of an nLGAD W6. By systematically evaluating the collected charge at bias voltages above V_{g1} , the evolution of the device thickness was extracted. The thickness Δz was determined directly from the width parameter of the rectangular box function in the fit model, which effectively defines the Full Width at Half Maximum of the active region. As summarized in Tab. II, a clear expansion of the active region is observed at higher voltages, agreeing with the support wafer depletion hypothesis.

C. Gain reduction on nLGADs

As for conventional LGADs, nLGADs also achieve moderate gain in the signal thanks to their n-type gain layer. In this type

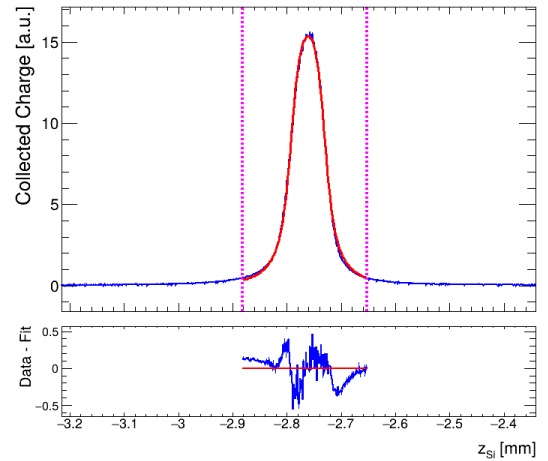


Figure 5: Example of the fit function (red line) applied to the collected charge profile (blue line) of the nLGAD W6 at $V_{bias} = 50$ V. The magenta dashed lines represent the boundaries allowed for the fit. The bottom panel displays the residuals (Data - Fit) to assess the fit quality.

of devices, a fraction of the carriers produce several secondary electron-hole pairs due to a very strong electric field inside the gain layer [19-21]. In order to study the different n-type gain layers, SPA measurements were performed with both red

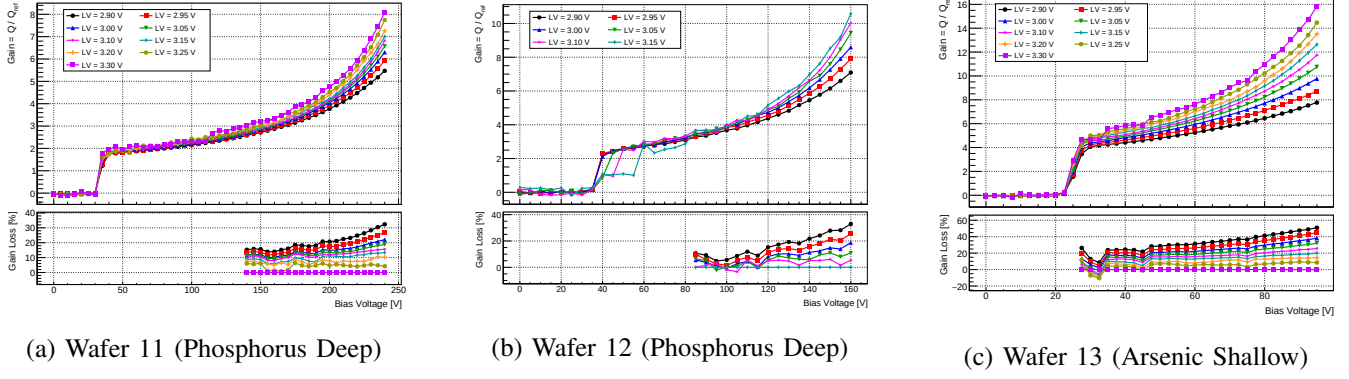


Figure 6: Gain reduction measurements using red SPA-TCT for different nLGAD designs. Top panels show the effective gain as a function of bias voltage for varying laser intensities (LV). Bottom panels display the relative gain loss referenced to the lowest intensity scan. All sensors exhibit strong gain suppression at higher injection densities, with Wafer 13 showing the most pronounced effect.

and infrared lasers, utilizing the experimental setup previously detailed in Sec. III.

The gain of an nLGAD is measured by comparing the collected charge of the desired detector with the reference PiNs,

$$\text{Gain} = \frac{Q(\text{nLGAD})}{Q(\text{PiN})}. \quad (1)$$

For estimating the gain, we first did a PiNs calibration. For this calibration, both red and infrared lasers were used at different laser intensities. To modulate the laser intensity in the setup, a variable optical attenuator is used, which further attenuates the laser while increasing the applied operation voltage.

Fig. 6 shows the TCT gain measurements using a $\lambda_{\text{red}} \approx 660$ nm wavelength at different laser intensities over a range of applied bias voltages for the nLGADs coming from wafers 11, 12 and 13, respectively. Gain suppression is observed for high values of laser intensities, where a screening of the electric field happens as the charge density increases significantly compared to lower values of the intensity. Comparing the different plots, it is observed that the gain reduction is not the same in the three studied detectors due to their fabrication differences. The wafer 13 nLGAD, featuring an arsenic-shallow gain layer, presents the highest gain reduction effect. In contrast, the nLGADs from wafers 11 and 12 show a very similar gain reduction because they share a common phosphorus-deep gain layer, differing only slightly in their implantation doses (0.96 for Wafer 11 versus 1.00 for Wafer 12).

Fig. 7 shows the infrared ($\lambda_{\text{IR}} \approx 1064$ nm) SPA-TCT gain measurements for the devices from wafers 11, 12, and 13. Notably, no gain reduction was observed despite applying higher laser intensities. This behavior is explained by the localized charge carrier density. In red SPA-TCT, electron-hole pairs are generated just a few microns below the surface, confined to a small spherical volume defined by the short penetration depth of the red light (~ 3.2 μm). This creates a

high charge density that screens the electric field and reduces the gain. In contrast, infrared light generates carriers uniformly throughout the active thickness. These charges are distributed across a much larger cylindrical volume, defined by the laser's FWHM (~ 18 μm) and the sensor's thickness (~ 55 μm). Consequently, the carrier density for the infrared laser is roughly 100 times lower, effectively avoiding the localized charge excess and the subsequent gain reduction.

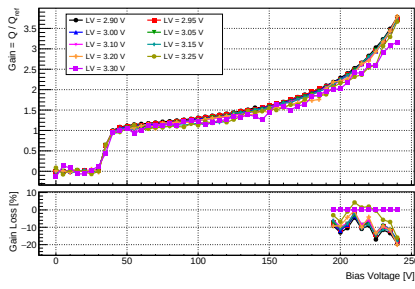
However, another type of measurement was carried out to study the gain reduction. This new measurement consisted on varying the z position of the laser in order to displace the laser from its focus point, as shown in Fig. 8. Although the infrared wavelength has not the same behavior as the red, if the laser is displaced from the focus point, a slight difference in the amount of generated charged carriers is expected, allowing to observe the gain reduction. The results present a gain loss of $\sim 5\%$ in the focus point ($z = 44.50$ mm) compared to the measurement less biased by this gain reduction mechanism ($z = 46.00$ mm).

V. CONCLUSION

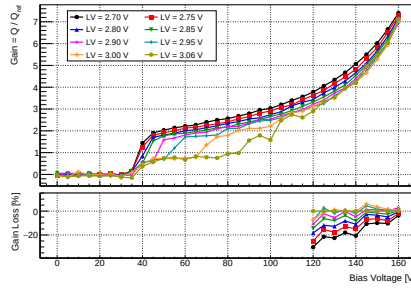
The studies presented in this work demonstrated a deep characterization of the nLGAD devices fabricated by FBK using different TCT, providing us a global understanding of this type of detectors. An analysis over a wide range in applied bias voltages using SPA and TPA measurements give clues on the behavior for bias voltages near and below the V_{gl} , about how the internal electric field varies and how this variation explains the different waveforms obtained during the measurements.

After increasing the voltage beyond the V_{gl} , the detector seems to be widening its active region, possibly starting to deplete the support wafer, giving a wider effective active region for low-penetrating particles.

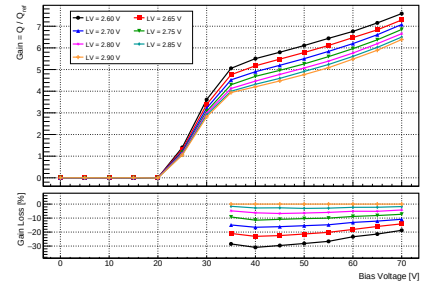
Finally, the gain reduction was studied on different nLGADs so as to understand the variations in the gain reduction mechanism among the different designs of the n -type gain layers and to see which type of gain layer provides a more



(a) Wafer 11 (Phosphorus Deep)



(a) Wafer 12 (Phosphorus Deep)



(b) Wafer 13 (Arsenic Shallow)

Figure 7: Gain and relative gain loss using infrared SPA-TCT for wafers 11, 12 and 13. Unlike the red laser measurements, no significant gain suppression is observed here due to the deeper penetration depth of IR light, which prevents the formation of a highly dense, localized charge plasma at the surface.

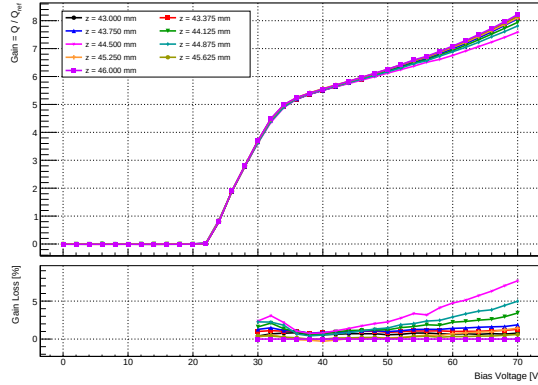


Figure 8: Gain suppression in the W13 nLGAD evaluated by varying the IR laser focal depth (z). Moving the laser out of focus slightly decreases the local charge density, revealing a relative gain loss of $\sim 5\%$ at the focal point ($z = 44.50$ mm) compared to a defocused position ($z = 46.00$ mm).

satisfactory result for future fabrication of these devices and their applications.

APPENDIX: ANALYTICAL DERIVATION OF THE FITTING FUNCTION

As introduced previously, the spatial response profile of the detector is modeled by convoluting the Lorentzian behavior of the laser energy deposition in TPA with the expected rectangular geometric profile of the sensor's active volume.

Mathematically, the active region of the detector is described by a rectangular function $\Pi(z)$ of width w centered at a coordinate z_0 . The spatial profile of the TPA laser deposition is represented by a Lorentzian distribution $L(z)$ with a half-width at half-maximum (HWHM) parameter γ . The resulting measured profile $f(z)$ is given by their convolution:

$$f(z) = A \int_{-\infty}^{+\infty} L(z-t) \Pi(t) dt + C, \quad (2)$$

where A is an overall amplitude scaling factor and C accounts for the experimental baseline offset.

By defining the top-hat function in terms of Heaviside step functions, $\Pi(t) = \Theta(t - z_0 + \frac{w}{2}) - \Theta(t - z_0 - \frac{w}{2})$, the integration boundaries are strictly confined to the physical limits of the active region. The convolution integral simplifies to:

$$f(z) = \frac{A}{\pi} \int_{z_0 - w/2}^{z_0 + w/2} \frac{\gamma}{(z-t)^2 + \gamma^2} dt + C. \quad (3)$$

Evaluating the integral at these limits yields the analytical expression used to fit the experimental data:

$$f(z) = \frac{A}{\pi} \left[\arctan\left(\frac{z - z_0 + w/2}{\gamma}\right) - \arctan\left(\frac{z - z_0 - w/2}{\gamma}\right) \right] + C. \quad (4)$$

ACKNOWLEDGMENT

I would like to first thank my supervisor Marcos for giving me the opportunity to join the Solid State Detectors group and teaching me all the physics behind this area of physics and showing me how to work in the laboratory with both TPA and SPA setups. Also the good times that we spent while measurements were done and all my time spent seeing how ROOT plots were done.

I would also like to thanks Luca for being my supervisor while Marcos was not able to be at CERN, for giving me support with the research and making me feel more comfortable inside the group.

Then, I would like also to thank Matteo, that, although not being my supervisor, he acted like one, helping me in every laboratory task and making the time doing the measurements better. Also, thanks for helping in the analysis of all the measurements, without your help it would not be possible.

I would also like to thank the entire SSD group for being so welcoming and supportive during my stay at CERN, and for including me in all the meetings. It has been a true pleasure working with such a wonderful team.

Finally, thanks to all the people I met during the Summer Student programme. It has been a pleasure to meet all of you.

I would also like to give a special mention to the Iberian team (winners of the CERN World Cup), I really enjoyed playing football with you. —

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